

2.5 Curvature_Guide

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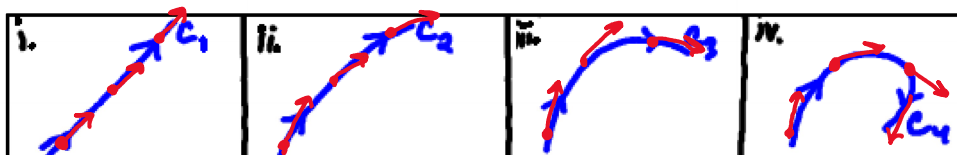
2.5
Curvature...

2.5 Curvature

We will now explore the notion of curvature. We begin by reviewing some terminology.

Definition: A parameterization $\mathbf{r}(t)$ is called smooth on an interval I if $\mathbf{r}'$ is continuous and $\mathbf{r}'(t) \neq \mathbf{0}$. A curve is smooth if it has a smooth parameterization. A smooth curve has no sharp corners or cusps.

Remark: For a smooth curve given by $\mathbf{r}(t)$ when the unit tangent vector, $\mathbf{T}(t)$, turns it does so continuously. When a smooth curve has sharp turns $\mathbf{T}(t)$ will turn quickly. When a curve turns quite slowly, $\mathbf{T}(t)$ will not turn quickly. Consider the pictures below for examples of this idea.



This leads to the following definition.

Definition: The curvature of a smooth curve is $\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\|$.

Theorem: The curvature of a smooth curve given by $\mathbf{r}(t)$ can be expressed as

$$1) \quad \kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}.$$

OR

$$2) \quad \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$$

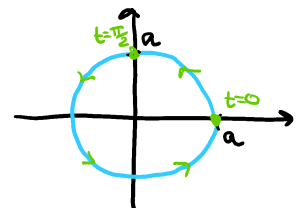
Proof: We will prove (2). A proof of (1) is available in Blackboard.

Note: Part (b) of this theorem will tend to be more useful since it only requires that we know the vector-valued function $\mathbf{r}(t)$, we don't need to compute the tangent vector $\mathbf{T}(t)$, and we can easily change the parameter if needed.

Example 1: Consider the circle given by $\mathbf{r}(t) = \langle a \cos(t), a \sin(t) \rangle$ where $0 \leq t < 2\pi$ and $a > 0$. Find the curvature of this circle.

$$\begin{aligned} \mathbf{r}'(t) \times \mathbf{r}''(t) &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -a \sin(t) & a \cos(t) & 0 \\ -a \cos(t) & -a \sin(t) & 0 \end{vmatrix} \\ &= \langle 0, 0, a^2 \sin^2(t) + a^2 \cos^2(t) \rangle = \langle 0, 0, a^2 \rangle \\ \text{So } \|\mathbf{r}'(t) \times \mathbf{r}''(t)\| &= \sqrt{0+0+a^4} = a^2 \\ \|\mathbf{r}'(t)\| &= \sqrt{a^2 \sin^2(t) + a^2 \cos^2(t) + 0} = \sqrt{a^2} = a \end{aligned} \quad \left. \vphantom{\begin{aligned} \mathbf{r}'(t) \times \mathbf{r}''(t) \\ \|\mathbf{r}'(t) \times \mathbf{r}''(t)\| \\ \|\mathbf{r}'(t)\| \end{aligned}} \right\} \text{ So } \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3} = \frac{a^2}{a^3} = \frac{1}{a}$$

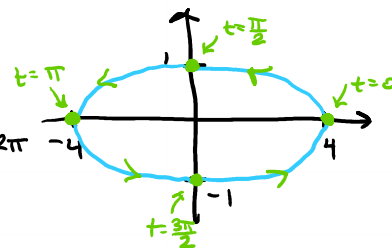
$\kappa(t) = \frac{1}{a}$



Example 2: Let C be the ellipse $x^2 + 16y^2 = 16$ oriented counterclockwise. Measure the curvature at each endpoint of its major and minor axes.

$$x^2 + 16y^2 = 16 \rightarrow \frac{x^2}{16} + \frac{y^2}{1} = 1$$

$$\text{Let } \vec{r}(t) = \langle 4\cos(t), 1\sin(t) \rangle \quad 0 \leq t \leq 2\pi$$



$$\vec{r}'(t) = \langle -4\sin(t), \cos(t), 0 \rangle$$

$$\vec{r}''(t) = \langle -4\cos(t), -\sin(t), 0 \rangle$$

$$\begin{aligned} \vec{r}'(t) \times \vec{r}''(t) &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -4\sin(t) & \cos(t) & 0 \\ -4\cos(t) & -\sin(t) & 0 \end{vmatrix} = \langle 0, 0, 4\sin^2(t) + 4\cos^2(t) \rangle \\ &= \langle 0, 0, 4 \rangle \end{aligned}$$

$$\text{So } \|\vec{r}'(t) \times \vec{r}''(t)\| = 4$$

$$\& \|\vec{r}'(t)\| = \sqrt{16\sin^2(t) + \cos^2(t)} = \sqrt{15\sin^2(t) + 1}$$

$$\text{Thus } K(t) = \frac{4}{\left(\sqrt{15\sin^2(t) + 1}\right)^3}$$

$$K(0) = K(\pi) = \frac{4}{1} = 4$$

$$K\left(\frac{\pi}{2}\right) = K\left(\frac{3\pi}{2}\right) = \frac{4}{4^3} = \frac{1}{16}$$